# Muhammed Muminul Hoque

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## Education

**Bachelor of Science in Computer Science & Engineering** GPA: 3.35/4.00
University of Dhaka, Bangladesh (3.58 in last 60 credits)
February 2019 – March 2024

## Research Experience

**Research Intern** November 2025 – Present
Under the supervision of *Professor Yanfu Zhang*
Department of Computer Science, William & Mary, Williamsburg, VA

- Developing a "Position-Aware" In-Context Learning (ICL) evaluation pipeline for long-context LLMs (Qwen2.5-7B) to mitigate performance degradation ("Context Rot") in multi-shot prompting.
- Engineering automated, large-scale inference pipelines via **vLLM** on NVIDIA RTX A6000 (48GB VRAM) to test whether ordering candidate examples using the "Lost in the Middle" attention bias yields zero-cost accuracy gains.
- Co-authored *DyV2X*, a dynamic graph learning framework for V2X cooperative trajectory prediction, submitted to IROS 2026 (**2nd author**).
- Designed a GRU history encoder across 6 CTDG architectures (TGN, TGAT, GraphMixer, DyGFormer, DyRep, JODIE) on the V2X-Seq-TFD dataset; achieved minADE = 0.952 m, minFDE = 1.664 m, MR = 24.7%.

## Publications

### Published

E. Hossain, P. R. Saha, **M. M. Hoque**, M. S. Islam, J. Al Faruk, and M. A. Hossain, **ECCE 2025**
***"Privacy Preserving and Secure Digital Identity Verification Mechanism: AES-Encrypted Smart NID with Two-Factor Protection Against Unauthorized Access,"***
in *Proc. 4$^{th}$ Int. Conf. on Electrical, Computer and Communication Engineering (ECCE)*,
CUET, Chattogram, Bangladesh, 2025, pp. 1–6.
doi:10.1109/ECCE64574.2025.11012996

P. R. Saha, Md. S. Islam, T. K. Saha, **M. M. Hoque**, A. Haque, E. Hossain, **STI 2025**
***"Quantum-Resistant Blockchain: Enhancing Blockchain Security with ROUND5, A Lattice-based Cryptosystem,"***
in *Proc. IEEE 7$^{th}$ Int. Conf. on Sustainable Technologies for Industry 5.0 (STI 2025)*.
doi:10.1109/STI69347.2025.11367567

- Implemented ROUND5 lattice-based cryptography to reduce post-quantum vulnerability; proposed an efficient key exchange protocol suitable for SMEs.

### Accepted

Jonayed Al Faruk, **Muhammed Muminul Hoque**, Md. Nawaz Shorif, Susmoy Paul, Md. Bodrul Islam, Joyonto Das, **BIM 2026**
***"Toward Accurate Urban Temperature Forecasting in Seoul: A Hybrid Ensemble of Bias-Corrected Machine and Deep Learning Models with Explainable AI,"***
in *Proc. 3$^{rd}$ Int. Conf. on Big Data, IoT and Machine Learning (BIM 2025)*, Springer Lecture Notes in Networks and Systems, Vol. 1, 2026, accepted for publication.

- Explored XGBoost, KNN, CNN, SVR, and Feedforward Neural Networks; applied SHAP and LIME for explainability of meteorological and topographic features.

**Muhammed Muminul Hoque**, Md. Masudur Rahman, Jonayed Al Faruk, A.K.M. Bahalul Haque, Mir Tasrif Ahmed, MD. Nayeem Pervez **ECCT 2026**
***"A Comparative Analysis of Traditional and Ensemble Machine Learning Models for Phishing Email Detection with Explainable AI."***
in *Proc. Int. Conf. on Electrical, Computer and Communication Technologies* ***(ECCT 2026)***, Dhaka International University, Bangladesh, May 7–9, 2026. To be published in a Taylor & Francis Book.

- Built a unified dataset of 290,404 labeled emails from 12 public corpora.
- Benchmarked classical vs. ensemble ML models with explainable AI (XAI); achieved ROC-AUC of 94.6%.

**Submitted & Under Review**

Yiran Ding, **Muhammed Muminul Hoque**, Maiqi Jiang, Yongming Qin, Sidi Lu, Yanfu Zhang **IROS 2026 (Under Review)**
***"DyV2X: Dynamic Interaction Graphs for V2X Cooperative Trajectory Prediction"***
*Submitted to IEEE/RSJ International Conference on Intelligent Robots and Systems* ***(IROS 2026)***, Pittsburgh, PA, Sept 27–Oct 1, 2026.

- Proposed a dynamic interaction graph framework that captures the temporal evolution of inter-agent interactions for V2X cooperative motion forecasting.
- Demonstrated consistent outperformance over prior methods, with particularly significant gains in safety-critical turning scenarios.

**Selected Research in Progress**

**M. M. Hoque**, Md. Masudur Rahman, A.K.M. Bahalul Haque. ***"ZEAL: A Zero-Shot Heterogeneous Ensemble Framework for Adaptive Anomaly Detection."*** *Target: IEEE T-IFS or similar Q1 venue.* **(In Prep)**

- Developed a multi-tier agentic pipeline utilizing heterogeneous model ensembles and reasoning escalation for zero-shot detection — zero training data required.
- Achieves 98.44% accuracy on 4,812 samples across two benchmarks, outperforming fine-tuned transformer baselines through advanced logical consensus.

**M. M. Hoque**, Maiqi Jiang, Y. Zhang, et al., ***"Optimizing In-Context Learning via Non-Linear Attentional Priors in Long-Context Windows."*** *Target: ACL 2026 or NAACL 2026.* **(In Prep)**

- Proposed an inference-only method using empirical attentional analysis to optimize example positioning in long-context ICL, significantly recovering performance lost in mid-context.
- Benchmarked on state-of-the-art LLMs (7B+) across 4k+ token context lengths, demonstrating consistent accuracy gains over naive relevance-based ordering.

## Project Work

**Aurora: Autonomous Academic Research Agent optimized for low-resource servers** 2026

- Engineered a multi-agent AI system optimized for heavily constrained edge hardware (1GB RAM), utilizing OpenRouter to dynamically route tasks between specialized LLMs (Hermes 405B, Qwen, Gemini Flash) with zero local memory overhead.
- Built an asynchronous Telegram interface featuring real-time token streaming, computer vision (document parsing), and automated synthesis of ArXiv research papers.
- Implemented a robust background task scheduler integrating multi-backend deadline tracking across local JSON, Notion APIs, and Google Calendar.
- Tools: Python, AsyncIO, python-telegram-bot, APScheduler, Gemini API.

**PeerTranslate: Verified Research Paper Translation for Every Language** 2026

- Open-source AI translation tool for research papers with a 4-pass back-translation verification pipeline (Translate → Back-Translate → Score → Re-translate), producing per-section confidence scores.

- Powered by Google Gemini with community-curated domain glossaries (ML, CS, general academic) to prevent mistranslation of technical jargon; supports multiple languages with open contributions.
- Tools: Python, FastAPI, Google Gemini API.
- Live demo: peertranslate.onrender.com

**Toxicity Analysis of 4chan's /pol/ Board Using OpenAI Moderation and Google Perspective APIs** 2025

- Developed an end-to-end pipeline to detect and analyze toxic speech patterns using OpenAI Moderation API and Google Perspective API for multi-dimensional toxicity scoring.
- Conducted ethical data collection, quantitative analysis, and visualization to identify trends in online discourse.
- Tools: Python, Pandas, Matplotlib.
- *Reviewed by Prof. Kai-Cheng Yang (Binghamton University).*

**Machine Learning Based Speech Emotion Recognizer** 2024

- Identifies and classifies emotions from speech, providing real-time emotional insights.
- Tools: JupyterLab, Python.
- Supervisor: Prof. Md. Alamgir Hossain, Professor (CSE) & Principal.

## Technical Skills

**Languages & Frameworks:** Python, C/C++, Bash, PyTorch

**Graph Neural Networks & Motion Forecasting:** TGN, TGAT, GraphMixer, DyGFormer, GCN, GAT, PyG, DGL, HiVT, DenseTNT, V2X-Graph, WOMD evaluation

**ML & Explainability:** Scikit-learn, SHAP, LIME, GNNExplainer, XGBoost, Random Forest, SVM

**Tools & Libraries:** Pandas, NumPy, Matplotlib, Git, GitHub, Overleaf, Google Colab

## Training Experiences & Certifications

**Cyber Security For Professionals**, Byte Capsule Feb 2023 – Aug 2023

- Six-month program covering threat analysis, risk management, network security, cryptography, and ethical hacking through hands-on exercises and expert-led lectures.

## Standardized Test Scores

**International English Language Testing System (IELTS)** 14 December 2024

- Overall: 6.5 Listening: 8.0 Reading: 6.0 Writing: 6.0 Speaking: 6.0

## Extra-curricular Activities, Awards, and Achievements

- MEC Computer Club (January 2023 – March 2024)
- 3rd Place at MEC CSE Carnival 2020 Programming Competition
- Batch Topper – 6-Month Self-learning Cybersecurity Training Program (Feb – Aug 2023)
- Champion of Invisible CTF 2023 (18 September 2023)
- Ranked in the Top 6% on TryHackMe
- Solved 95+ challenges in PicoCTF, including 50+ medium-level problems across cryptography, reverse engineering, and web security

## References

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